FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
CONCEPT RECAPITULATION TEST – I
PAPER –1
TEST DATE: 26-04-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. A, D

Sol. Let r_1 be radius of conical conductor at left end, then radius at distance "a" from left end

$r_2 = r_1 + ba$ [b = constant]

$$j = \frac{i}{A} = \sigma E \Rightarrow E = \frac{i}{A\sigma}$$

$$E \propto \frac{1}{A}\left[= \frac{1}{\pi(r_1 + ab)^2}\right]$$

$\therefore$ Thus, E – a curve is wrong.

$$H = \frac{I^2\rho}{A}$$

$$H \propto \frac{I}{A}\left[= \frac{1}{\pi(r_1 + ab)^2}\right]$$

Thus, (H – a) curve is correct.

$$\frac{H}{E} = \frac{i^2\rho}{A}\left(\frac{A}{i\rho}\right) = i = \text{constant}$$

Thus, (H – E) curve is a straight line passing through origin.

$$dv = -\vec{E}.d\vec{x} = -\int_0^a \frac{\rho i dx}{\pi(r_1 + ab)^2}$$

$$v = \frac{\rho i}{\pi b}\left[\frac{ab}{r_1 + ab}\right]$$

Thus (V – a) is wrong.

2. A, C

Sol. $e_{ind} = \frac{d\phi}{dt} = \pi a^2\alpha; i = \frac{\pi a^2\alpha}{\rho(2\pi a)} = \frac{a\alpha}{2\rho}$

For More Material Join: @JEEAdvanced_2024

3. A, D

Sol. $\frac{1}{f_{eff}} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{30}$

$\frac{1}{f_{eff}} = \frac{1}{v} - \frac{1}{u}$

$\Rightarrow$ 1/v = 0

$\Rightarrow$ $v = \infty$

$f = \frac{R}{2(\mu - 1)}$

$\Rightarrow$ R = 20 cm

For $\mu_{surrounding}$ = 9/8

$f = \frac{20}{2\left[\frac{3\times 8}{2\times 9} - 1\right]} = \frac{10}{\left(\frac{4}{3} - 1\right)}$

f = 30 cm

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$\frac{1}{30} = \frac{1}{v} - \frac{1}{-30}$

$v = \infty$

4. A, D

Sol. Some field lines must go to infinity also from Q, and some on conductor. Hence total flux leaving Q is greater than that entering conductor.
Hence A
Also, since potential of conductor is zero, hence it can't have positive charge anywhere on surface else it will emit some field lines to infinity making its potential non-zero.
Hence D

5. B, D

Sol. By gas equation
PV = nRT

$4.155\times 10^5 \times 70 \times 10^{-3} = n \times 8.31 \times 350$

So, n = 10
Now by first law of thermodynamics
$\Delta Q = \Delta U + \Delta W$
ΔU = Power $\times$ Time – (8077.5) = 6000 – 8077.5 = - 2077.5 Joule
$nC_V\Delta T = -2077.5$

So, $C_V = \frac{-2077.5}{10\times(-10)} \Rightarrow C_V = 20.775$

$\frac{f}{2}R = 20.775$

$f = \frac{2\times 20.775}{8.31} = 5$

6. A, B, C

Sol. $V_1 - \omega_1 R_1 = \omega_2 R_2$

$\Rightarrow$ $V_1 = \omega_1 R_1 + \omega_2 R_2$

Differentiating, we get

$a_1 = \alpha_1 R_1 + \alpha_2 R_2$

$V_1, a_1 :\rightarrow$ Speed and tangential acceleration of 'P'

For More Material Join: @JEEAdvanced_2024

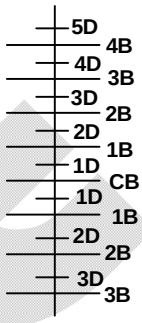
7. A

Sol. $\vec{dF} = i\left(\vec{d\ell} \times \vec{B}\right)$

$\vec{\tau} = i\vec{A} \times \vec{B}$

8. A

Sol. By using $(\mu\text{-}1)t = n\lambda$, we can find value of n, that is order of the fringe produced at P, if that particular strip has been placed over any of the slit. If two strips are used in conjunction (over each other), path difference due to each is added to get net path difference created. If two strips are used over different slits, their path differences are subtracted to get net path difference.



Now, $n_1 = \dfrac{(\mu_1 - 1)t_1}{\lambda} = 5$

$n_2 = 4.5$

and $n_3 = 0.5$

For (I), order of the fringe is 4.5 i.e. 5th dark

For (II), net order is 5 – 0.5 = 4.5

i.e. fifth dark

For (III) net order is 5 – (0.5 + 4.5) = 0

i.e. it is central bright again at P.

For (IV) net order is (5+0.5) – (4.5) = 1

i.e. first bright

9. A

Sol. $u = 40\cos 37^\circ \hat{i} + 40\sin 37^\circ \hat{j},\ \vec{v} = 40\cos 37^\circ \hat{i} + \left(40\sin 37^\circ - gt\right)\hat{j}$

(I) $\vec{u}.\vec{v} = 0\ (32)^2 + 24\ (24 - 10t)$

(II) 0 = 20 –10 t, t = 2 sec (at highest point)

(III) $-36 = 24t - 5t^2 \Rightarrow 5t^2 - 24t - 36 = 0$, t = 6 sec

(IV) $-\left(40\sin 37^\circ - gt\right) = 40\cos 37^\circ$

$-(24 - 10t, = 32\ t = \dfrac{56}{10} = 5.6$ sec)

10. B

Sol. $q_B = -q\left(1 - \dfrac{1}{K}\right)$

$q_A = +q$

$q_C = +q\left(1 - \dfrac{1}{K}\right)$

$q_D = -q$

Section – B

11. 1.05

Sol. Energy incident on cesium metal $= \dfrac{1240}{400} = 3.1$eV

$K_{mx} = 3.1 - 1.9 = 1.2$ eV

(k_{mx} = maximum kinetic energy of photoelectron from the metal cesium)

For More Material Join: @JEEAdvanced_2024

So kinetic energy of photoelectron
$0 \le k \le k_{mx}$.

So, $k = \dfrac{\frac{\sigma e}{2\epsilon_0} \times \ell + k_{mx}}{\frac{\sigma e}{2\epsilon_0}\ell} = \dfrac{23.8\text{ eV}}{22.6\text{ eV}} = 1.053$

12. 2.39

Sol. $E = C_1C_2\cos\omega_0 t + \dfrac{C_1C_3}{2}\cos(\omega_0+\omega)t + \dfrac{C_1C_3}{2}\cos(\omega_0-\omega)t$

Of the three components, the highest frequency component will liberate the electrons with maximum kinetic energy

$(K.E)_{max} + \phi = \dfrac{h}{2\pi}(\omega_0+\omega) \Rightarrow \phi$ = 2.39 eV.

13. 3.00

Sol. $T_r = \dfrac{(2r+1)(2r^2+2r-1)}{(r^2-1)^2(r^2+2r)^2} = \dfrac{(2r+1)(2r^2+2r-1)}{(r^2-1)^2((r+1)^2-1)^2}$

$\therefore T_r = \left[\dfrac{1}{(r^2-1)^2} - \dfrac{1}{((r+1)^2-1)^2}\right]$

$\therefore K = \lim\limits_{n\to\infty}\sum\left(\left(\dfrac{1}{(2^2-1)^2} - \dfrac{1}{(3^2-1)^2}\right) + \left(\dfrac{1}{(3^2-1)^2} - \dfrac{1}{(4^2-1)^2}\right) + ... + \left(\dfrac{1}{(n^2-1)^2} - \dfrac{1}{((n+1)^2-1)^2}\right)\right)$

$\therefore k = \dfrac{1}{9}$

14. 0.15

Sol. Energy of incident photons in eV is given as

$$E = \frac{12431}{1800}\text{eV}$$

As work function of metal is 2eV, the maximum kinetic energy of ejected electrons is

$$KE_{max} = E - \phi$$

$\Rightarrow$ $KE_{max} = 6.9 - 2\text{eV}$

$\Rightarrow$ $KE_{max} = 4.9\text{eV}$

If ν_{max} be the speed of fastest electrons then we have

$$\frac{1}{2}m\nu_{max}^2 = 4.9\times1.6\times10^{-19}\text{ joule}$$

$\Rightarrow$ $\nu_{max} = \sqrt{\dfrac{2\times4.9\times1.6\times10^{-19}}{9.1\times10^{-31}}}$

$\Rightarrow$ $\nu_{max} = 1.31\times10^6$ m/s

When an electron with this speed enters a uniform magnetic field normally it follows a circular path whose radius can be given by

$$r = \frac{mv}{qB} \qquad [\text{As } qvB = \frac{mv^2}{r}]$$

$\Rightarrow$ $r = \dfrac{9.1\times10^{-31}\times1.31\times10^6}{1.6\times10^{-19}\times5\times10^{-5}}$

$\Rightarrow$ $r = 0.149$ m

For More Material Join: @JEEAdvanced_2024

15. 1.66

Sol. If central maxima is formed at point P on the screen, then

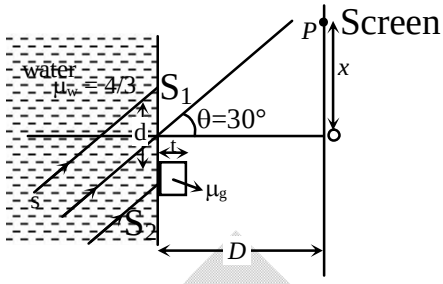
Path difference = $S_2P + (\mu_g - 1)t - S_1P - \mu_w d \sin 30° = 0$

$\Rightarrow$

$$S_2P - S_1P = \frac{xd}{D} = \mu_w d \sin 30° - (\mu_g - 1) \times t$$

$\Rightarrow$

$$x = -1.66 \text{ cm}$$



16. 654.00

Sol. $Mg - T = Ma$ and $TR = \left(\frac{1}{2}MR^2\right)\alpha = \frac{MR^2}{2} \times \frac{a}{R}$

$\therefore \quad Mg - M\frac{a}{2} = Ma$

$a = \frac{2}{3}g$ = 654 cm/s^2

17. 8.00

Sol. $\because$ Elements of determinant can be 1 or -1

$\therefore$ First two places of first row can be filled in 2 × 2 =4 ways and last place can be filled in only one way to product be 1

$\therefore$ First row can be made in 4 ways

Similarly second row can be made in 4 ways and third row can be made in 1 ways only

$\therefore$ m = 16

$$\because |P| = \begin{vmatrix} 3 & -2 & 3 \\ 2 & -2 & 3 \\ 0 & -1 & 1 \end{vmatrix} = 1$$

$$\because \text{adj}(\text{adj}P) = |P|^{n-2} P = P \Rightarrow \text{tr}(\text{adj}(\text{adj}P)) = \text{tr}(P) = 2 = n.$$

$$\therefore \frac{m}{n} = \frac{16}{2} = 8$$

18. 100.00

Sol. $B = -\frac{P}{\Delta V / V}$ or $P = -B.\frac{\Delta V}{V}$

$\therefore \quad P = 9.8 \times 10^8 \times \left(\frac{0.1}{100}\right)$

or $\quad h\rho g = 9.8 \times 10^8 \times \left(\frac{0.1}{100}\right)$

$\therefore \quad h = \frac{9.8 \times 10^8}{1000 \times 9.8} \times \frac{0.1}{100}$ m = 100 m

For More Material Join: @JEEAdvanced_2024

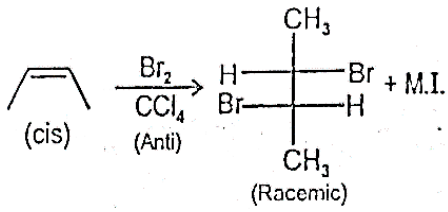
Chemistry

PART – II

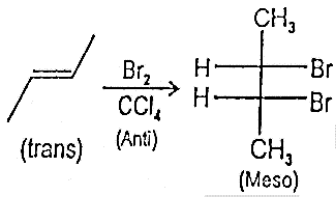
Section – A

19. C, D

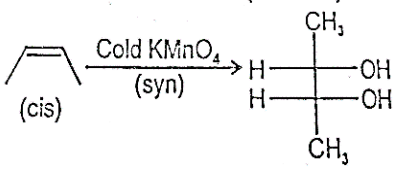
Sol. (A)



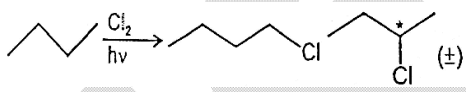
(B)



(C)



(D)



20. B, C, D

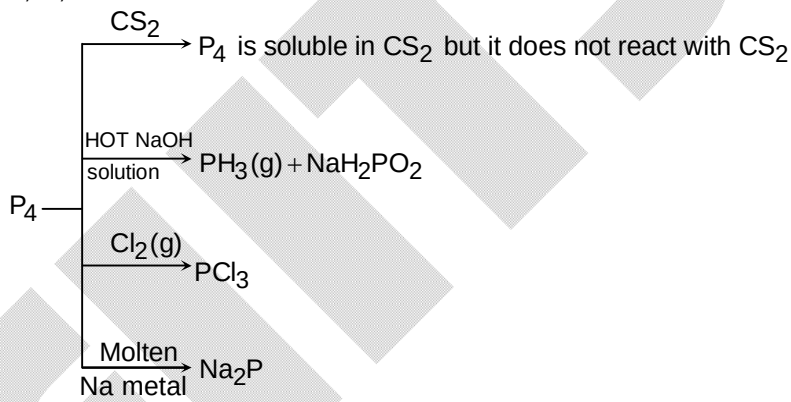
Sol. Molecular weight of C_6H_5COOH in $CH_3COCH_3 = \frac{K_b \times W_{solute} \times 1000}{\Delta T_b \times W_{solvent}} = \frac{1.70 \times 1.22 \times 1000}{0.17 \times 100} = 122$

Molecular weight of C_6H_5COOH in $C_6H_6 = \frac{2.6 \times 1.22 \times 1000}{0.13 \times 100} = 244$

Hence, molecular weight of C_6H_5COOH in both the solvent is not equal. C_6H_5COOH dimerise in C_6H_6 and remains in monomeric form in acetone.

21. B, C, D

Sol.



22. A, D

Sol. According to Bohr model angular momentum = $nh/2\pi$, n = 3 thus A is incorrect
Number of electrons is nucleus = 0, thus D is incorrect

23. A, B, C, D

Sol. (A) End othermic raction temperature ↑ equilibrium shift in forward direction so degree of dissociation ↑

(B) $1+\alpha = \frac{D}{d}$ $\quad T \uparrow \alpha \uparrow d \downarrow$

(C) $P_{N_2O_4} \downarrow P_{NO_2} \downarrow$ when helium gas added under constant pressure.

(D) $P_{N_2O_4} \uparrow$ when N_2O_4 added

For More Material Join: @JEEAdvanced_2024

24. A, C, D

Sol.

Complex compounds	Type of structural iso.
(A) $\left[Ir(ONO)_3(H_2O)_3\right]$	linkage isomerism
(B) $\left[Ir(NO_3)(H_2O)_5\right]^{2+}$	None
(C) $\left[Co(ox)(H_2O)_3Br\right].H_2O$	Hydrate isomerism
(D) $\left[PdBr(NH_3)_4(SO_4)\right]Cl$	Ionization isomerism

25. B

Sol.

$NO_2 + H_2O \longrightarrow HNO_3 + NO(g)$

$XeF_2 + HF \longrightarrow \left[XeF\right]^+\left[F^-.....H-F\right]$

$OCl_2 + H_2O \longrightarrow 2HClO$

$BF_3 + HF \longrightarrow H^+\left[BF_4\right]^-$

26. B

Sol.

(I) CH_3COOH (J, ×) and CH_3CH_2OH (M, ✓) (NaOI)

(II) CH_3COOH (J, ×) and $CH_3-\overset{O}{\overset{\|}{C}}-CH_3$ (K, ✓) (2, 4-DNP)

(III) $CH_3-\overset{O}{\overset{\|}{C}}-CH_3$ (K, ✓) and CH_3CH_2OH (M, ×) ($NaHSO_3$)

(IV) CH_4 (L, ×) and CH_3CH_2OH (M, ✓) (Na-metal)

27. D

Sol.

(I) $\Delta T_f = 0.1\times 2 = 0 - T_f = 273 - 0.2 = 272.8K$

(II) $\pi = 0.1(2)\times 0.0821\times 298 = 4.98\,atm$

(III) $\Delta T_b = 2.5\left[1+\left(\frac{1}{2}-1\right)\right]0.1 = \frac{0.25}{2} = 0.125$

(IV) $RLVP = \frac{P^o - P}{P^o} \simeq \frac{in}{N} = \frac{0.1}{\frac{1000}{18}} = 18\times 10^{-4}$

$\therefore \%\,RLVP = 100\times\frac{P^o-P}{P^o} = 18\times 10^{-2}$ or 0.18

$\underset{0.1-0.1\alpha}{C_6H_2OH} + H_2O \rightleftharpoons \underset{0.1\alpha}{C_6H_5O^-}(aq) + \underset{0.1\alpha}{H_3O^+}(aq)$

For More Material Join: @JEEAdvanced_2024

$$K_a = 10^{-10} = \frac{\left[C_6H_5O^-\right]\left[H_3O^+\right]}{\left[C_6H_5OH\right]} = 0.1\alpha^2$$

$$\alpha = 10^{-3}, i = \frac{0.1\alpha + 0.1}{0.1} = \alpha + 1 = 1 + 10^{-3} \simeq 1$$

28. A

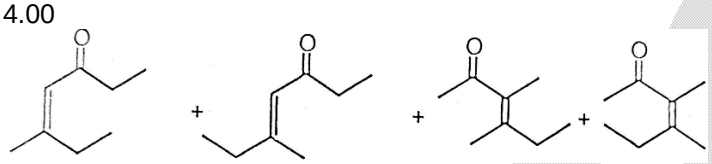
Sol.

Ionic Compound	voids Occupied
NaCl	Octahedral voids occupied
ZnS	50% tetrahedral voids occupied
CsCl	Cubic void occupied
CaF_2	100% tetrahedral voids occupied

Section – B

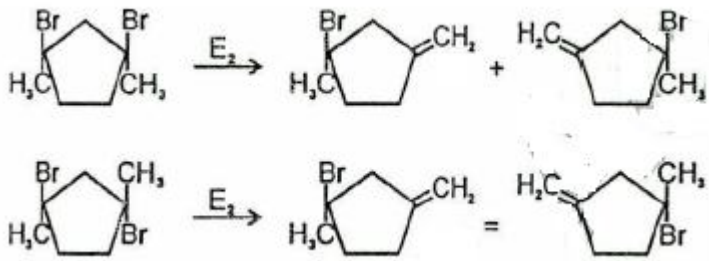
29. 4.00

Sol.



30. 3.00

Sol.



31. 6160.00

Sol. Litres of gasoline used = $\frac{50}{20} = 2.5L$

Mass of gasoline used = $2.5 \times 1000 \times 0.8$ g = 2000 g

Mass of carbon consumed = 0.84×2000 g = 1680.

Moles of carbon consumed = $\frac{1680}{12}$

Moles of CO_2 formed = $\frac{1680}{12}$

Mass of CO_2 produced = $\frac{1680}{12} \times 44$

= 6160

32. 2.38

Sol. $2Mn + 4\,HCl + O_2 \rightarrow 2\,MnCl_2 + 2\,H_2O$

$$E = E^o - \frac{RT}{4F}\ell n\frac{\left[MnCl_2\right]^2}{PO_2\left(HCl\right)^4}$$

For More Material Join: @JEEAdvanced_2024

$= 1.229 + 1.185 - \frac{0.06}{4} \log\left[\frac{10^{-6}}{0.25\times10^{-8}}\right]$

$= 2.375$

33. 6.00

Sol. (ii, iii, iv, v, vii, viii) are more basic than aniline

34. 30.00

Sol. Let initial no. of tritium atoms = x

Mass of T_2O + mass of H_2O = 0.0009

$\Rightarrow \frac{x}{N_A}\left[22+\frac{18}{8\times10^{-18}}\right] = 0.0009$

$\Rightarrow x = \frac{0.0009}{18}\times8\times10^{-18}\times6\times10^{23} = 240$

Atoms of tritium left after 3 half lives = $\frac{240}{8} = 30$

35. 66.00

Sol. No. of Si per unit cell = 8

Let radius of Si atom = R

$2R = \frac{\sqrt{3}a}{4} \Rightarrow a = \frac{8R}{\sqrt{3}}$

% of vacant space = $100 - \frac{8\times\frac{4}{3}\pi R^3}{a^3}$

$= 100 - \frac{8\times4\times\pi\times3\sqrt{3}}{3\times8^3}$

= 66%

36. 72.00

Sol. $O_3 \rightarrow \frac{3}{2}O_2$

$K = 5\times10^{-4}$

$-\frac{d(O_3)}{dt} = 5\times10^{-4}\left(2\times10^{-8}\right)^2$ mol $L^{-1}s^{-1}$

$= 5\times10^{-4}\times\left(2\times10^{-8}\right)^2\times6\times10^{23}\times\frac{1}{1000}\times60$

= 7200 molecules mL^{-1} min^{-1}

For More Material Join: @JEEAdvanced_2024

Mathematics

PART – III

Section – A

37. B, C

Sol. $z = r(\cos\theta + i\sin\theta)$

$$\left|5r\cos\theta + i5r\sin\theta + \frac{\cos\theta}{r} - \frac{i\sin\theta}{r}\right| = 3$$

$$\left(5r\cos\theta + \frac{\cos\theta}{r}\right)^2 + \left(5r\sin\theta - \frac{\sin\theta}{r}\right)^2 = 9$$

$$25r^2 + \frac{1}{r^2} + 10\left(\cos^2\theta - \sin^2\theta\right) = 9$$

$$\cos^2\theta = \frac{9 - \left(25r^2 + \frac{1}{r^2}\right)}{10}$$

$$2\cos^2\theta - 1 \le -\frac{1}{10}$$

$$\cos^2\theta \le \frac{9}{20}$$

38. A, B, C

Sol. In right triangle

$$\text{Inradius} = \frac{1}{2}(\text{base} + \text{height} + \text{hypotaneous})$$

$$\text{Circum radius} = \frac{\text{hypotanens}}{2}$$

In isosceles right angled Δ, sides are is ratio 1 : 1 : $\sqrt{2}$, So required ratio is $\sqrt{2} + 1$
In right triangle having angles is AP is $\sqrt{3} + 1$
In regular hexogen required ratio is 2 :$\sqrt{3}$
In equalled Δ, radii of incircle, circumcircle and ex circle are in ratio 1 : 2: 3

39. A, B, C

Sol. $2^x + 2^y = 2^{x+y} \Rightarrow 2^{-x} + 2^{-y} = 1$

$$-2^{-y}.\ln 2\frac{dy}{dx} + 2^{-1}\ln 2(-1) = 0 \Rightarrow \frac{dy}{dx} = -\frac{2^y}{2^x}$$

$$1 + \frac{2^y}{2^x} = 2^y \qquad \Rightarrow 1 - 2^y = \frac{-2^y}{2^x}$$

$$\frac{dy}{dx} = 1 - 2^y$$

Also $\frac{2^x}{2^y} + 1 = 2^x \qquad \Rightarrow -\frac{2^y}{2^x} = \frac{1}{1-2^x}$

$$\frac{dy}{dx} = \frac{1}{1-2^x}$$

40. A

Sol. Let Q be the centre of the given circle and R be the centre of the smallest circle touching the given circle externally than

For More Material Join: @JEEAdvanced_2024

$AQ = 3 + \sqrt{2}$

$PQ = 3\sqrt{2} + 4$

$\Rightarrow 3\sqrt{2} + 4 = 3 + 2\sqrt{2} + r + r\sqrt{2}$ $\left(\because \angle APQ = \frac{\pi}{4}\right)$

$\Rightarrow (\sqrt{2}+1) = r(\sqrt{2}+1) \Rightarrow r = 1.$

41. C, D

Sol. Let $A = \theta + \frac{5\pi}{18} \Rightarrow A \in \left[\frac{\pi}{6}, \frac{\pi}{4}\right]$

$f(A) = \tan A + \cos A + \cot A \ \forall \ A \in \left[\frac{\pi}{6}, \frac{\pi}{4}\right]$

Hence $I_1 = \frac{11}{2\sqrt{3}}, I_2 = \frac{2\sqrt{2}+1}{\sqrt{2}}$

42. A, B

Sol. Let P (2, -1) goes 2 units along x + y = 1 upto A and 5 units along x – 2y = 4 up to B

Slope of PA = -1= tan135°, slope of PB = $\frac{1}{2}$ = tanθ

$\therefore \sin\theta = \frac{1}{\sqrt{5}}, \cos\theta = \frac{2}{\sqrt{5}}$

$\therefore$ A ≡ (x_1 + rcos135°, y_1 = rsin135°) = (2 + 2x $-\frac{1}{\sqrt{2}}$, -1 + 2 × $\frac{1}{\sqrt{2}}$)

= (2 - √2, √2 – 1)

B ≡ (x_1 + rcosθ, y_1 + rsinθ) = (2 + 5 × $\frac{2}{\sqrt{5}}$, -1 + $\frac{5}{\sqrt{5}}$) = (2√5 + 2, √5 – 1)

43. B

Sol. (I) We have, $\alpha = \frac{z - \bar{\omega}}{16 + z\bar{\omega}} \Rightarrow \bar{\alpha} = \frac{\bar{z} - \omega}{16 + \bar{z}\omega}$

But $z\bar{z} = \omega\bar{\omega} = 16$

So, $\bar{\alpha} = \frac{\frac{16}{z} - \frac{16}{\bar{\omega}}}{16 + \left(\frac{16}{z}\right)\left(\frac{16}{\bar{\omega}}\right)} = -\alpha$

$\Rightarrow \alpha + \bar{\alpha} = 0 \Rightarrow \mathrm{Re}(\alpha) = 0$

(II) $x = \frac{x^3}{x^2} = \left(\frac{2+11i}{3+4i}\right) \times \frac{(3-4i)}{(3-4i)} = (2+i)$

$\therefore \ p = 2, q = 1$

$\Rightarrow (p+q) = 2+1 = 3$

(III) We have, $\bar{z} = iz^2$; Put z = x + iy

$\Rightarrow x - iy = i(x^2 - y^2 + 2ixy)$

$\therefore$ On solving, we get

$z = 0 + i0, i, \frac{\sqrt{3}}{2} - \frac{i}{2}, \frac{-\sqrt{3}}{2} - \frac{i}{2}$

$\Rightarrow$ 4 solutions.

For More Material Join: @JEEAdvanced_2024

(IV) $|3z+9-7i| = |(3z+6-3i)+(3+4i)|$

$\le |3z+6-3i| + |3-4i|$

$= 3|z+2-i| + 5$

$= 20$

44. A

Sol. (I) Equally like mutually exclusive and exhaustive cases to have exactly 2H in first three and exactly 2H in last three, when the coin is thrown 5 times, are

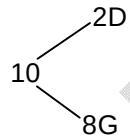
$\therefore$ n(S) = 5

$$\begin{bmatrix} H & H & T & H & H \\ H & T & H & H & T \\ H & T & H & T & H \\ T & H & H & H & T \\ T & H & H & T & H \end{bmatrix}$$

Out of these five, 4 cases favours the event, that 3rd flip is H.

Hence $P(E) = \frac{4}{5}$

(II) E : atmost 3 are drawn

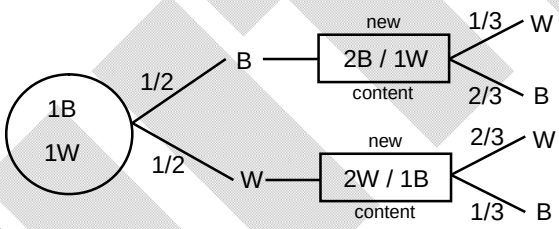


$P(E) = P(G \text{ or } DG \text{ or } DDG)$

$= \frac{8}{10} + \left(\frac{2}{10}\right)\left(\frac{8}{9}\right) + \left(\frac{2}{10}\right)\left(\frac{1}{9}\right)(1)$

$= \frac{8}{10} + \frac{16}{90} + \frac{2}{90} = \frac{72+16+2}{90} = 1$

(III)

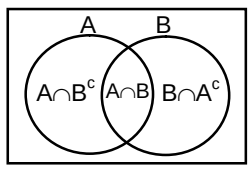


A : 1st drawn is white

B = at least one of the two balls drawn was white

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{2}\cdot\frac{2}{3} + \frac{1}{2}\cdot\frac{1}{3}}{\frac{1}{2}\cdot\frac{1}{3} + \frac{1}{2}\cdot\frac{2}{3} + \frac{1}{2}\cdot\frac{1}{3}} = \frac{\frac{1}{3}+\frac{1}{6}}{\frac{1}{6}+\frac{1}{3}+\frac{1}{6}} = \frac{\frac{1}{2}}{\frac{4}{6}} = \frac{1}{2}\cdot\frac{6}{4} = \frac{3}{4}$$

(IV)



$P(A \cap B^c) = P(A) - P(A \cap B)$

$P(A \cap B^c)$ will be minimum if $P(A \cap B)$ is maximum

But $P(A \cap B)_{max} = 0.3$ (think)

$\therefore$ $(A \cap B^c)_{min} = 0.7 - 0.3$

$= 0.4 = \frac{4}{10} = \frac{2}{5}$

For More Material Join: @JEEAdvanced_2024

45. C

Sol. (I) $\int_0^x t.f(2x-t)dt = \frac{1}{2}\tan^{-1}(x^2)$

Put $2x - t = y$

$\Rightarrow \quad dt = -dy$

Hence $\int_x^{2x}(2x-y)f(y)dy = 2x\int_x^{2x}f(y)dy - \int_x^{2x}yf(y)dy$

$\therefore 2x\int_x^{2x}f(y)dy - \int_x^{2x}yf(y)dy = \frac{1}{2}\tan^{-1}(x^2)$

Differentiating both sides

$2\left[x(2f(2x)-f(x)) + \int_x^{2x}f(y)dy\right] - [2.2x.f(2x) - xf(x)] = \frac{2x}{2(1+x^4)}$

$4xf(2x) - 2xf(x) + 2\int_x^{2x}f(y)dy - 4xf(2x) + xf(x) = \frac{x}{1+x^4}$

$2\int_x^{2x}f(y)dy = xf(x) + \frac{x}{1+x^4}$

Put $x = 1,\ 2\int_1^2 f(y)dy = f(1) + \frac{1}{2} = \frac{3}{2}$

$\int_1^2 f(x)dx = \frac{3}{4}$

$\Rightarrow \quad 4\int_1^2 f(x)dx = 3 \quad \Rightarrow (\mathbf{S})$

(II) Let $I = \int_1^{2009} \underbrace{((x-1)(x-2)...)(x-1004)...(x-2007)(x-2008)(x-2009)dx}_{\text{2009 terms=f(x)say}}$

Using King

$I = \int_1^{2009}(2009-x)(2008-x)(2007-x)...(2004-x)(2003-x)...(1-x)dx$

$= (-1)^{2009}\int_1^{2009}f(x)dx = -I$

$\therefore \quad 2I = 0$

$\Rightarrow \quad I = 0 \Rightarrow \mathbf{(P)}$

(III) $I = \int_0^{\ln 10}\frac{e^x\sqrt{e^x-1}}{e^x+8}dx$

Let $e^x - 1 = t^2$

$\Rightarrow e^x dx = 2t\,dt$

$I = \int_0^3 \frac{t^2dt}{t^2+9} = \left[\int_0^3 dt - 9\int_0^3\frac{dt}{t^2+9}\right]$

$= 2\left[3 - \frac{9}{3}\tan^{-1}\frac{t}{3}\Big|_0^3\right] = 6 - 6\left(\frac{\pi}{4}\right) = 6 - \frac{3\pi}{2}$

$\Rightarrow a = 6$ and $b = \frac{3}{2}$

$\Rightarrow a/b = 4 \Rightarrow (T)$

(IV) $G'(x) = \frac{g'(x)}{g^2(x)}$

$G'(2) = -\frac{g'(2)}{g^2(2)} = -\frac{g'(2)}{16}$

For More Material Join: @JEEAdvanced_2024

Now $(fog)(x) = x$

$f'(g(x)).g'(x) = 1$

$\Rightarrow \quad f'(g(2)).g'(2) = 1$

$\Rightarrow \quad f'(4).g'(2) = 1$

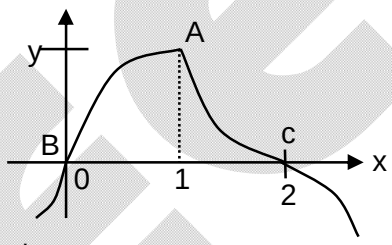
$\Rightarrow \quad g'(2) = 16$

From (ii), $G'(2) = -\frac{16}{16} = -1$

$\Rightarrow \quad (G'(2))^2 = 1$

46. D

Sol. (I) A, B, C are the 3 critical points of y = f(x).
f''(x) = 0 for x = 2 and fails to exists at x = 0.



(II) x = ¼ and 2. Make a quadratic in $\log_2 x$ and interpret the result.

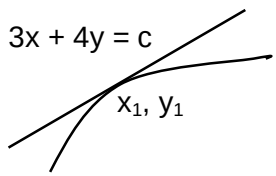
(III) $\frac{dy}{dx} = -1 + 2x_1^3 = -\frac{3}{4} \Rightarrow x_1 = \frac{1}{2}$

$\Rightarrow \frac{1}{32} = \frac{1}{2} + y_1$ or $y_1 = -\frac{15}{32} \Rightarrow c = -\frac{-3}{8}$

(IV) $f'(x) = 2x^3 - 3x + 1$ this is always positive in (1, 2).

$\therefore$ Increasing in [1, 2]

$\therefore$ f(2) will be the greatest value.



Section – B

47. 12.00

Sol. $\sum \frac{a_i}{a_j}$, $i, j \in \{1, 2, 3, 4\}$, $i \neq j$ has a total of $4 \times 4 - 4 = 12$ terms i.e. 6 pairs, each pair of the type $\left(\frac{a_i}{a_j} + \frac{a_j}{a_i}\right)$.

Now $\left(\frac{a_i}{a_j} + \frac{a_j}{a_i}\right) \geq 2 \qquad \left(\text{as } \frac{a_i}{a_j} > 0\right)$

$\Rightarrow \sum \frac{a_i}{a_j} \geq 2 \times 6 \Rightarrow \sum \frac{a_i}{a_j} \geq 12$.

48. 32.00

Sol. $x^3 - px^2 + qx - 8 = 0$

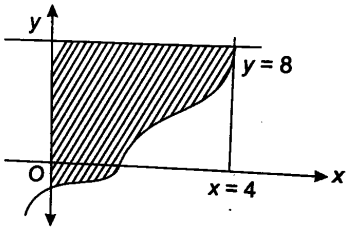
abc = 8 ($\because$ a, b, c are roots of the equation)

$\because \{\sqrt{n}\} = 0 \Rightarrow$ a, b, and c are even numbers.

$\Rightarrow N = 3^2.2^2.5^2$ and $f(x) = (x-2)^3$

For More Material Join: @JEEAdvanced_2024

$$A = \int_0^4 \left[8 - (x-2)^3\right] dx = (8x)_0^4 - \left(\frac{(x-2)^4}{4}\right)_0^4 = 32$$



49. 25.00

Sol. $\because {}^{30}C_r \cdot {}^{20}C_0 + {}^{30}C_{r-1} \cdot {}^{20}C_1 + ... + {}^{30}C_0 \cdot {}^{20}C_r$

= coefficient of x^r in $(1 + x)^{30}(1 + x)^{20}$

= coefficient of x^r in $(1+x)^{50} = {}^{50}C_r$

$\because {}^{50}C_r$ is maximum

$\therefore \quad r = \frac{50}{2} = 25$

50. 47.00

Sol. S(2, 3) is circumcentre $\Rightarrow$ SA = SB = SC

$\Rightarrow \quad \alpha = -\frac{13}{5}, \beta = 7 \Rightarrow$ B is $\left(-\frac{13}{5}, \frac{26}{5}\right)$, C is (7, 4) $\Rightarrow$ BC is x + 8y = 39, p + q = 47

51. 100.00

Sol. Given, $f(x) = \sqrt{-\log_2 \left|\frac{2}{\pi}\sin^{-1}\left(\frac{x+3}{5}\right)\right|}$

$\Rightarrow \log_2 \left|\frac{2}{\pi}\sin^{-1}\left(\frac{x+3}{5}\right)\right| \le 0$

$\Rightarrow 0 < \left|\frac{2}{\pi}\sin^{-1}\left(\frac{x+3}{5}\right)\right| \le 1 \ (\text{Double} - \text{inequality})$

$\Rightarrow -1 \le \frac{2}{\pi}\sin^{-1}\left(\frac{x+3}{5}\right) \le 1$

$\Rightarrow -\frac{\pi}{2} \le \sin^{-1}\left(\frac{x+3}{5}\right) \le \frac{\pi}{2}$

$\Rightarrow -1 \le \frac{x+3}{5} \le 1 \Rightarrow -5 \le x+3 \le 5$

$\Rightarrow -8 \le x \le 2 \qquad ...(1)$

Also, $\left|\frac{2}{\pi}\sin^{-1}\left(\frac{x+3}{5}\right)\right| > 0$

$\Rightarrow x \ne -3 \qquad ...(2)$

$\therefore (1) \cap (2)$

$\Rightarrow$ Domain of f(x) = $x \in [-8, 2] - \{-3\}$

(i) As, a, b $\in$ I and a, b $\in$ domain f(x), so, a & b $\in$(–8, –7, –6, –5, –4, –2, –1, 0, 1, 2}

Hence, the number ordered pairs (a, b) is = 10 × 10 = 100 Ans.

For More Material Join: @JEEAdvanced_2024

52. 11.00

Sol. Given $f(x)f'(-x) = f(-x)f'(x)$

$$\Rightarrow \frac{f'(x)}{f(x)} = \frac{f'(-x)}{f(-x)}$$

Integrating both sides

$$\ln(f(x)) = -\ln(f(-x)) + c$$

$$\ln(f(x)f(-x)) = c \Rightarrow f(x)f(-x) = e^c = c^1$$

but $f(0) = 3$

$$f(0)^2 = c \qquad \Rightarrow c = 9$$

$$f(x)f'(x) = 9$$

53. 16.00

Sol. $\left|\sqrt{3}\cos x - \sin x\right| = 2 \qquad \Rightarrow \cos\left(x - \frac{\pi}{16}\right) = 1 \text{ or } -1$

$$x = \frac{11\pi}{6}, \frac{23\pi}{6}, \frac{5\pi}{6}, \frac{17\pi}{6}$$

Hence four values

54. 30.00

Sol. $$P\left(\frac{A_3}{E_2}\right) = \frac{\frac{3}{10}\times\frac{1}{3}}{\frac{2}{10}\times\frac{1}{2} + \frac{3}{10}\times\frac{1}{3} + \frac{4}{10}\times\frac{1}{4}} \times \frac{1}{3}$$

For More Material Join: @JEEAdvanced_2024